

## EDA --> Exploratory Data Analysis

EDA is foundation of machine learning EDA technique will helps to convert raw data to clean data

1. Variable Identification
2. Univariate Analysis
3. Bivariate Analysis
4. Imputation Technique (Transformer)
5. Missing Value treatment
6. Outlier Treatment
7. New Variable Creation

### Convert Raw Data To Clean Data

```
import pandas as pd
```

```
pd.__version__
```

```
'2.3.3'
```

```
emp = pd.read_excel(r"C:\Users\HP\Downloads\Rawdata.xlsx")
```

```
emp
```

|   | Name   | Domain         | Age      | Location  | Salary   | Exp     |  |
|---|--------|----------------|----------|-----------|----------|---------|--|
| 0 | Mike   | Datascience#\$ | 34 years | Mumbai    | 5^00#0   | 2+      |  |
| 1 | Teddy^ | Testing        | 45' yr   | Bangalore | 10%%000  | <3      |  |
| 2 | Uma#r  | Dataanalyst^^# | NaN      | NaN       | 1\$5%000 | 4> yrs  |  |
| 3 | Jane   | Ana^^lytics    | NaN      | Hyderbad  | 2000^0   | NaN     |  |
| 4 | Uttam* | Statistics     | 67-yr    | NaN       | 30000-   | 5+ year |  |
| 5 | Kim    | NLP            | 55yr     | Delhi     | 6000^\$0 | 10+     |  |

```
id(emp)
```

```
1826792090192
```

```
emp.columns
```

```
Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
emp.shape
```

```
(6, 6)
```

```
emp.head()
```

|   | Name   | Domain         | Age      | Location  | Salary   | Exp     |
|---|--------|----------------|----------|-----------|----------|---------|
| 0 | Mike   | Datascience#   | 34 years | Mumbai    | 5^00#0   | 2+      |
| 1 | Teddy^ | Testing        | 45' yr   | Bangalore | 10%%000  | <3      |
| 2 | Uma#r  | Dataanalyst^^# | NaN      | NaN       | 1\$5%000 | 4> yrs  |
| 3 | Jane   | Ana^^lytics    | NaN      | Hyderbad  | 2000^0   | NaN     |
| 4 | Uttam* | Statistics     | 67-yr    | NaN       | 30000-   | 5+ year |

emp.tail()

|   | Name   | Domain         | Age    | Location  | Salary   | Exp     |
|---|--------|----------------|--------|-----------|----------|---------|
| 1 | Teddy^ | Testing        | 45' yr | Bangalore | 10%%000  | <3      |
| 2 | Uma#r  | Dataanalyst^^# | NaN    | NaN       | 1\$5%000 | 4> yrs  |
| 3 | Jane   | Ana^^lytics    | NaN    | Hyderbad  | 2000^0   | NaN     |
| 4 | Uttam* | Statistics     | 67-yr  | NaN       | 30000-   | 5+ year |
| 5 | Kim    | NLP            | 55yr   | Delhi     | 6000^\$0 | 10+     |

emp.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0    Name        6 non-null      object
1    Domain       6 non-null      object
2    Age          4 non-null      object
3    Location     4 non-null      object
4    Salary       6 non-null      object
5    Exp          5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

emp

|   | Name   | Domain         | Age      | Location  | Salary   | Exp     |
|---|--------|----------------|----------|-----------|----------|---------|
| 0 | Mike   | Datascience#   | 34 years | Mumbai    | 5^00#0   | 2+      |
| 1 | Teddy^ | Testing        | 45' yr   | Bangalore | 10%%000  | <3      |
| 2 | Uma#r  | Dataanalyst^^# | NaN      | NaN       | 1\$5%000 | 4> yrs  |
| 3 | Jane   | Ana^^lytics    | NaN      | Hyderbad  | 2000^0   | NaN     |
| 4 | Uttam* | Statistics     | 67-yr    | NaN       | 30000-   | 5+ year |
| 5 | Kim    | NLP            | 55yr     | Delhi     | 6000^\$0 | 10+     |

emp.isnull()

|   | Name  | Domain | Age   | Location | Salary | Exp   |
|---|-------|--------|-------|----------|--------|-------|
| 0 | False | False  | False | False    | False  | False |
| 1 | False | False  | False | False    | False  | False |
| 2 | False | False  | True  | True     | False  | False |
| 3 | False | False  | True  | False    | False  | True  |
| 4 | False | False  | False | True     | False  | False |
| 5 | False | False  | False | False    | False  | False |

```
emp.isna()
```

|   | Name  | Domain | Age   | Location | Salary | Exp   |
|---|-------|--------|-------|----------|--------|-------|
| 0 | False | False  | False | False    | False  | False |
| 1 | False | False  | False | False    | False  | False |
| 2 | False | False  | True  | True     | False  | False |
| 3 | False | False  | True  | False    | False  | True  |
| 4 | False | False  | False | True     | False  | False |
| 5 | False | False  | False | False    | False  | False |

```
emp.isnull().sum()
```

```
Name      0
Domain     0
Age        2
Location   2
Salary     0
Exp        1
dtype: int64
```

```
emp.columns
```

```
Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
emp.dtypes
```

```
Name      object
Domain     object
Age        object
Location   object
Salary     object
Exp        object
dtype: object
```

```
emp['Name']
```

```
0      Mike
1  Teddy^
2  Uma#r
3      Jane
```

```

4    Uttam*
5      Kim
Name: Name, dtype: object

```

```
emp.describe()
```

|               | Name | Domain         | Age      | Location | Salary | Exp |
|---------------|------|----------------|----------|----------|--------|-----|
| <b>count</b>  | 6    | 6              | 4        | 4        | 6      | 5   |
| <b>unique</b> | 6    | 6              | 4        | 4        | 6      | 5   |
| <b>top</b>    | Mike | Datascience#\$ | 34 years | Mumbai   | 5^00#0 | 2+  |
| <b>freq</b>   | 1    | 1              | 1        | 1        | 1      | 1   |

Cleaning the data of Categorical coloumn

```
emp['Name'] = emp['Name'].str.replace(r'\W', '', regex = True)
```

```
emp
```

|          | Name  | Domain         | Age      | Location  | Salary   | Exp     |
|----------|-------|----------------|----------|-----------|----------|---------|
| <b>0</b> | Mike  | Datascience#\$ | 34 years | Mumbai    | 5^00#0   | 2+      |
| <b>1</b> | Teddy | Testing        | 45' yr   | Bangalore | 10%%000  | <3      |
| <b>2</b> | Umar  | Dataanalyst^^# | NaN      | NaN       | 1\$5%000 | 4> yrs  |
| <b>3</b> | Jane  | Ana^^lytics    | NaN      | Hyderbad  | 2000^0   | NaN     |
| <b>4</b> | Uttam | Statistics     | 67-yr    | NaN       | 30000-   | 5+ year |
| <b>5</b> | Kim   | NLP            | 55yr     | Delhi     | 6000^\$0 | 10+     |

```
emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex = True)
```

```
emp['Domain']
```

```

0    Datascience
1         Testing
2    Dataanalyst
3         Analytics
4     Statistics
5             NLP
Name: Domain, dtype: object

```

```
emp['Location'] = emp['Location'].str.replace(r'\W', '' , regex = True)
```

```
emp['Location']
```

```

0        Mumbai
1    Bangalore
2          NaN
3    Hyderbad
4          NaN
5        Delhi
Name: Location, dtype: object

```

```
emp['Age'] = emp['Age'].str.extract(r'(\d+)') # d+ is digit match digit only.
```

```
emp['Age']
```

```
0    34
```

```
1    45
```

```
2    NaN
```

```
3    NaN
```

```
4    67
```

```
5    55
```

```
Name: Age, dtype: object
```

```
emp
```

|   | Name  | Domain      | Age | Location  | Salary    | Exp     |
|---|-------|-------------|-----|-----------|-----------|---------|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5^00#0    | 2+      |
| 1 | Teddy | Testing     | 45  | Bangalore | 10%%000   | <3      |
| 2 | Umar  | Dataanalyst | NaN | NaN       | 1\$5%000  | 4> yrs  |
| 3 | Jane  | Analytics   | NaN | Hyderbad  | 2000^0    | NaN     |
| 4 | Uttam | Statistics  | 67  | NaN       | 30000-    | 5+ year |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000^\$0 | 10+     |

```
#Cleaning of Numerical Data
```

```
emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex = True)
```

```
emp['Salary']
```

```
0    5000
```

```
1   10000
```

```
2   15000
```

```
3   20000
```

```
4   30000
```

```
5   60000
```

```
Name: Salary, dtype: object
```

```
emp['Exp'] = emp['Exp'].str.extract(r'(\d+)') # d+ is digit match digit only.
```

```
emp['Exp']
```

```
0     2
```

```
1     3
```

```
2     4
```

```
3    NaN
```

```
4     5
```

```
5    10
```

```
Name: Exp, dtype: object
```

```
emp
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | NaN | NaN       | 15000  | 4   |
| 3 | Jane  | Analytics   | NaN | Hyderbad  | 20000  | NaN |
| 4 | Uttam | Statistics  | 67  | NaN       | 30000  | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |

Missing value treatment

```
import numpy as np
```

```
clean_data = emp.copy() # Here created copy of clean data
clean_data
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | NaN | NaN       | 15000  | 4   |
| 3 | Jane  | Analytics   | NaN | Hyderbad  | 20000  | NaN |
| 4 | Uttam | Statistics  | 67  | NaN       | 30000  | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |

```
clean_data.isna().sum()
```

```
Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

```
clean_data['Age'] =
    clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'])))
```

```
clean_data['Age']
```

```
0      34
1      45
2    50.25
3    50.25
4      67
5      55
```

```
Name: Age, dtype: object
```

```
clean_data['Location'] =
    clean_data['Location'].fillna(clean_data['Location'].mode()[0])
```

```
clean_data['Location']
```

```
0      Mumbai
1    Bangalore
2    Bangalore
3    Hyderbad
4    Bangalore
5        Delhi
Name: Location, dtype: object
```

```
clean_data['Exp'] =
    clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'])))
```

```
clean_data['Exp']
```

```
0      2
1      3
2      4
3    4.8
4      5
5     10
Name: Exp, dtype: object
```

```
clean_data
```

|   | Name  | Domain      | Age   | Location  | Salary | Exp |
|---|-------|-------------|-------|-----------|--------|-----|
| 0 | Mike  | Datascience | 34    | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45    | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | 50.25 | Bangalore | 15000  | 4   |
| 3 | Jane  | Analytics   | 50.25 | Hyderbad  | 20000  | 4.8 |
| 4 | Uttam | Statistics  | 67    | Bangalore | 30000  | 5   |
| 5 | Kim   | NLP         | 55    | Delhi     | 60000  | 10  |

Now changing Datatypes as per coloumn names. To Do So we use ASTYPE: Convert system inbuilt data type to user define datatype.

```
type(clean_data)
```

```
pandas.core.frame.DataFrame
```

```
id(clean_data)
```

```
1826791842208
```

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null     object
```

```

1   Domain      6 non-null    object
2   Age         6 non-null    object
3   Location    6 non-null    object
4   Salary      6 non-null    object
5   Exp         6 non-null    object

```

```
dtypes: object(6)
```

```
memory usage: 420.0+ bytes
```

```
clean_data.columns
```

```
Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
clean_data['Age'] = clean_data['Age'].astype(int)
```

```
clean_data['Exp'] = clean_data['Exp'].astype(int)
```

```
clean_data['Salary'] = clean_data['Salary'].astype(int)
```

```
clean_data['Name'] = clean_data['Name'].astype('category')
```

```
clean_data['Domain'] = clean_data['Domain'].astype('category')
```

```
clean_data['Location'] = clean_data['Location'].astype('category')
```

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 6 entries, 0 to 5
```

```
Data columns (total 6 columns):
```

```

#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      category
1   Domain       6 non-null      category
2   Age          6 non-null      int64
3   Location     6 non-null      category
4   Salary       6 non-null      int64
5   Exp          6 non-null      int64

```

```
dtypes: category(3), int64(3)
```

```
memory usage: 938.0 bytes
```

```
clean_data
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 15000  | 4   |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 20000  | 4   |
| 4 | Uttam | Statistics  | 67  | Bangalore | 30000  | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |

```
clean_data.to_csv('clean_data.csv')
```

```
import os
```

```
os.getcwd()
```

```
'C:\\Users\\HP'
```

```
clean_data
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 15000  | 4   |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 20000  | 4   |
| 4 | Uttam | Statistics  | 67  | Bangalore | 30000  | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |

## EDA TECHNIQUE LETS APPLY

```
import matplotlib.pyplot as plt # visualization
import seaborn as sns
```

```
import matplotlib.pyplot as plt # visualization
import seaborn as sns
```

```
clean_data['Salary']
```

```
0    5000
1   10000
2   15000
3   20000
4   30000
5   60000
```

```
Name: Salary, dtype: int64
```

```
vis1 = sns.distplot(clean_data['Salary'])
```

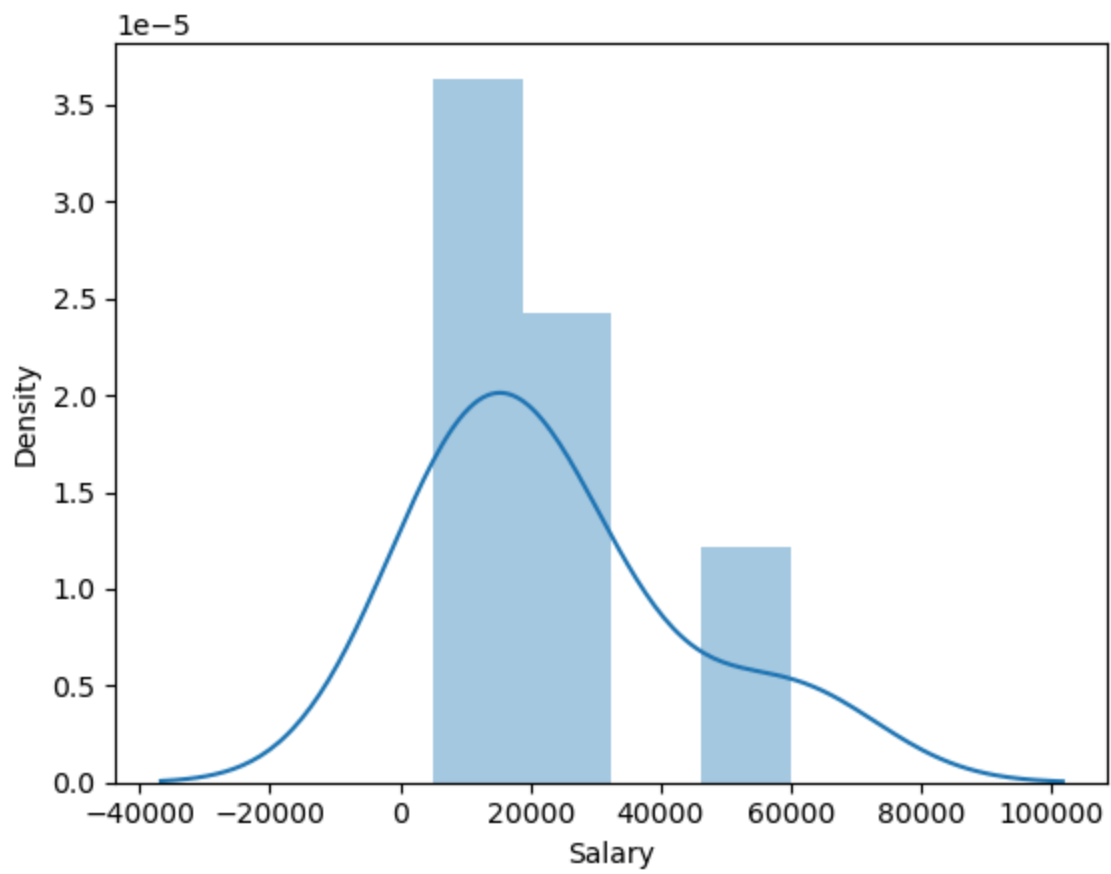
```
C:\Users\HP\AppData\Local\Temp\ipykernel_16340\3692155897.py:1: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

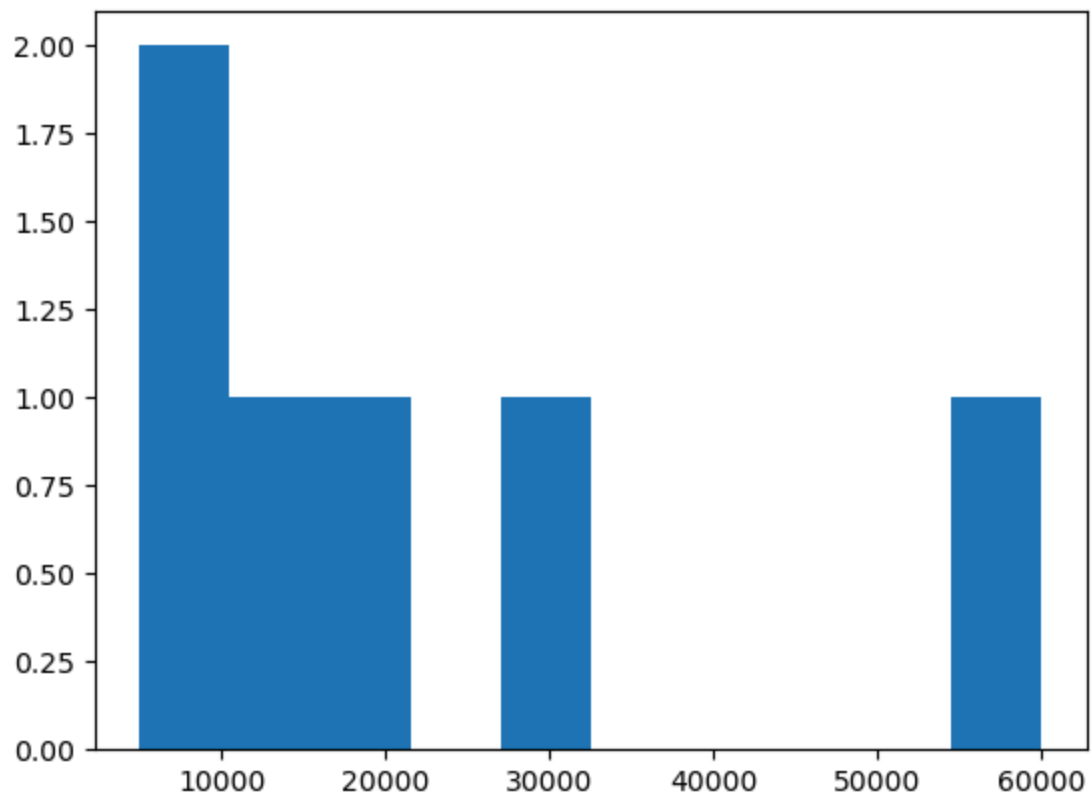
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

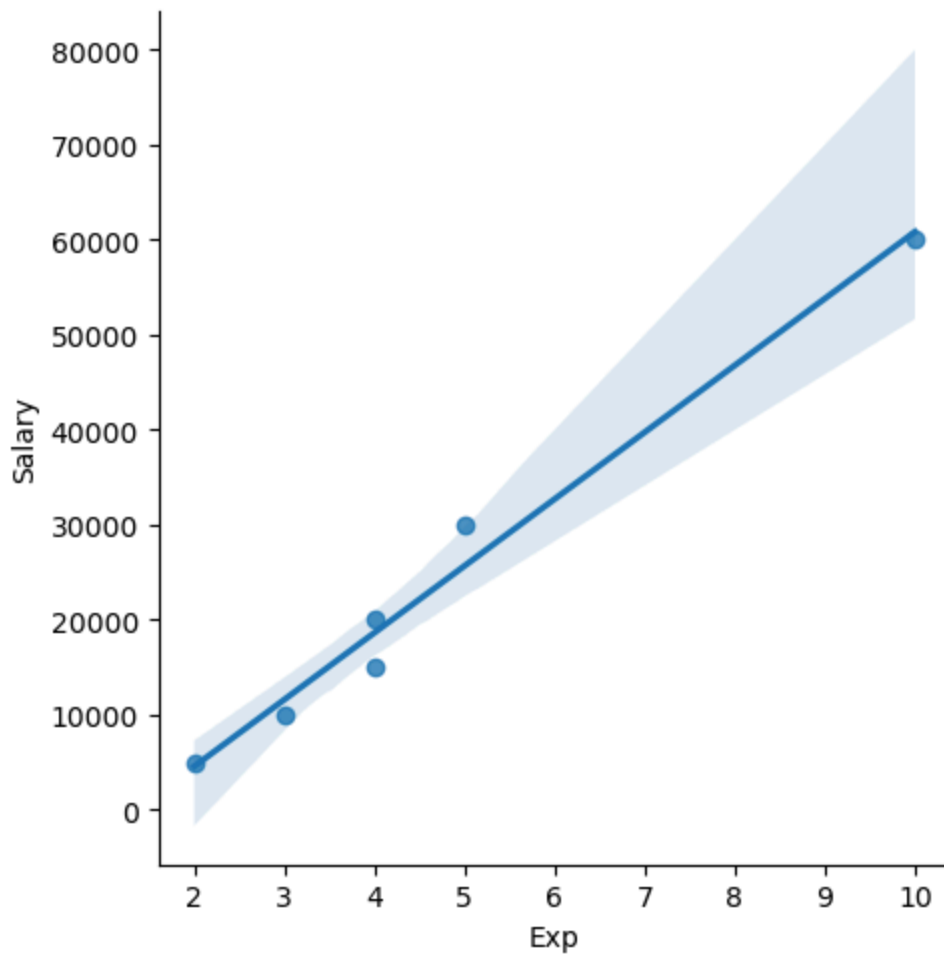
```
vis1 = sns.distplot(clean_data['Salary'])
```



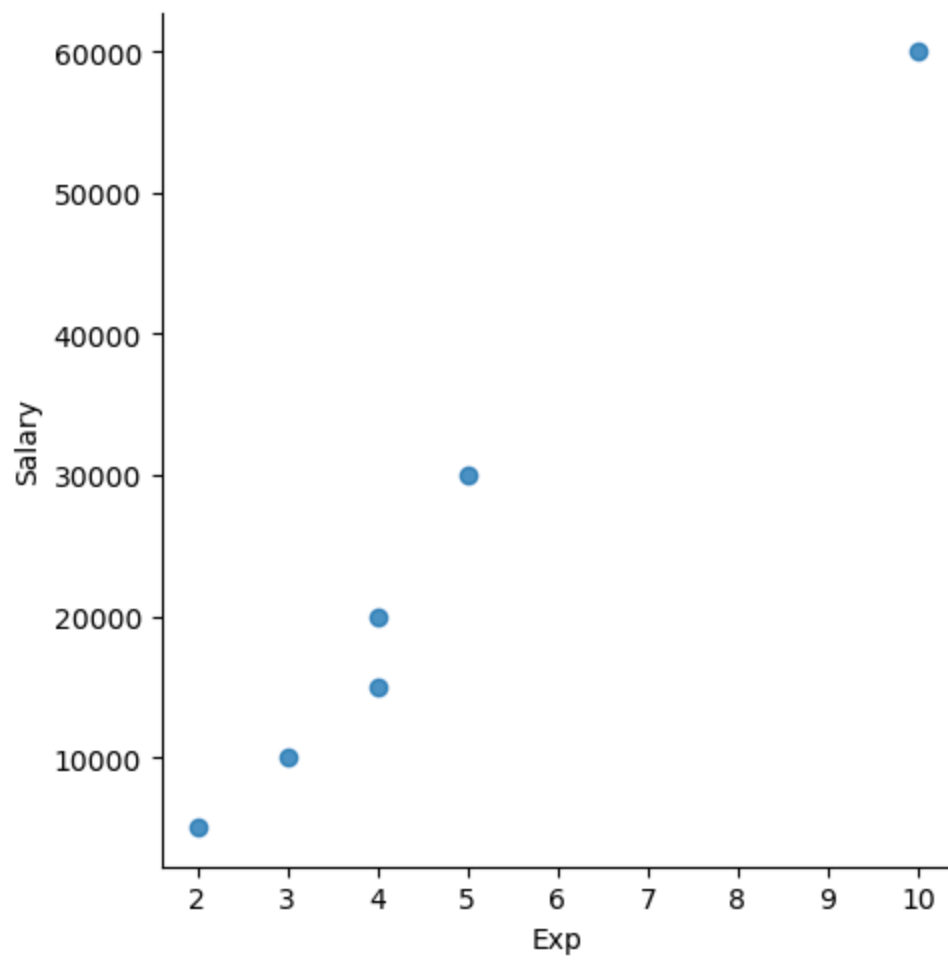
```
vis2 = plt.hist(clean_data['Salary'])
```



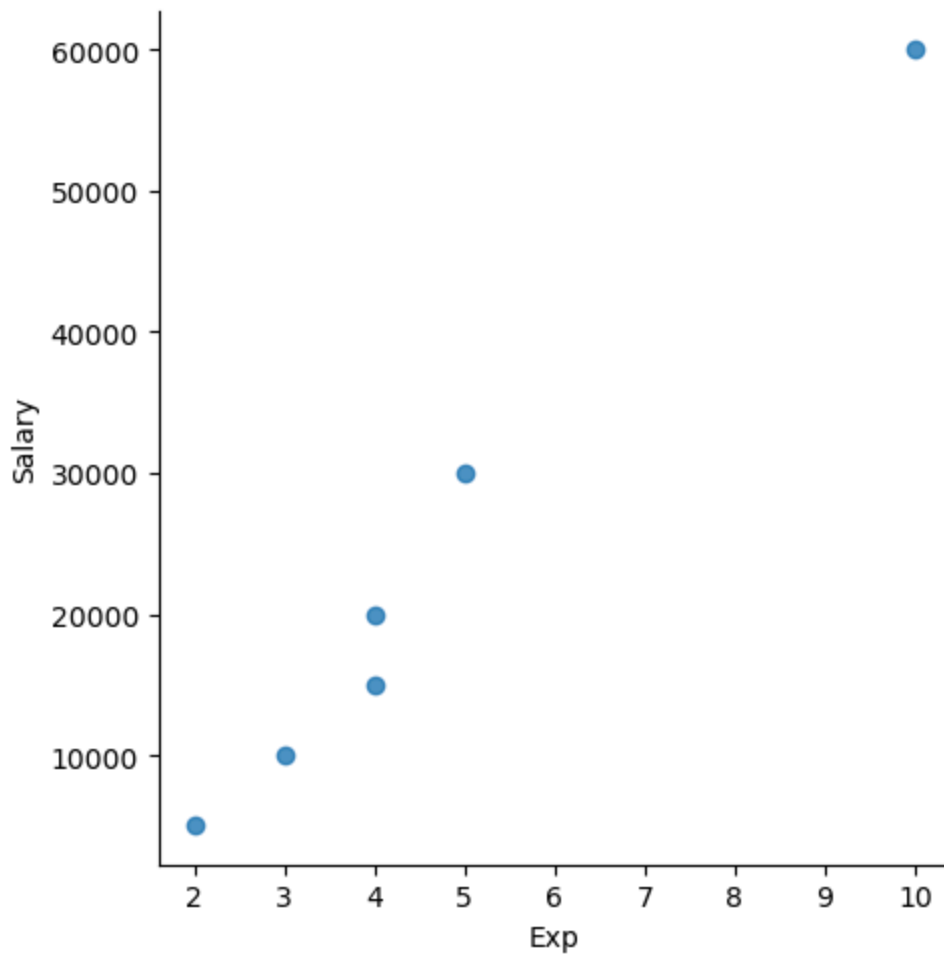
```
vis4 = sns.lmplot(data=clean_data, x = 'Exp', y='Salary')
```



```
vis5 = sns.lmplot(data=clean_data,x = 'Exp', y='Salary', fit_reg = False)
```



```
vis5 = sns.lmplot(data=clean_data,x = 'Exp', y='Salary', fit_reg = False)
```



```
clean_data[:]
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |  |
|---|-------|-------------|-----|-----------|--------|-----|--|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |  |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |  |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 15000  | 4   |  |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 20000  | 4   |  |
| 4 | Uttam | Statistics  | 67  | Bangalore | 30000  | 5   |  |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |  |

```
clean_data[0:6:2]
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |  |
|---|-------|-------------|-----|-----------|--------|-----|--|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |  |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 15000  | 4   |  |
| 4 | Uttam | Statistics  | 67  | Bangalore | 30000  | 5   |  |

```
clean_data[::-1]
```

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |
| 4 | Uttam | Statistics  | 67  | Bangalore | 30000  | 5   |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 20000  | 4   |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 15000  | 4   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |

`clean_data.columns`

`Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')`

`X_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]`

`X_iv`

|   | Name  | Domain      | Age | Location  | Exp |
|---|-------|-------------|-----|-----------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 3   |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 4   |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 4   |
| 4 | Uttam | Statistics  | 67  | Bangalore | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 10  |

`y_dv = clean_data[['Salary']]`

`y_dv`

|   | Salary |
|---|--------|
| 0 | 5000   |
| 1 | 10000  |
| 2 | 15000  |
| 3 | 20000  |
| 4 | 30000  |
| 5 | 60000  |

`emp`

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | NaN | NaN       | 15000  | 4   |

|   | Name  | Domain     | Age | Location | Salary | Exp |
|---|-------|------------|-----|----------|--------|-----|
| 3 | Jane  | Analytics  | NaN | Hyderbad | 20000  | NaN |
| 4 | Uttam | Statistics | 67  | NaN      | 30000  | 5   |
| 5 | Kim   | NLP        | 55  | Delhi    | 60000  | 10  |

clean\_data

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 15000  | 4   |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 20000  | 4   |
| 4 | Uttam | Statistics  | 67  | Bangalore | 30000  | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 60000  | 10  |

X\_iv

|   | Name  | Domain      | Age | Location  | Exp |
|---|-------|-------------|-----|-----------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 3   |
| 2 | Umar  | Dataanalyst | 50  | Bangalore | 4   |
| 3 | Jane  | Analytics   | 50  | Hyderbad  | 4   |
| 4 | Uttam | Statistics  | 67  | Bangalore | 5   |
| 5 | Kim   | NLP         | 55  | Delhi     | 10  |

y\_dv

|   | Salary |
|---|--------|
| 0 | 5000   |
| 1 | 10000  |
| 2 | 15000  |
| 3 | 20000  |
| 4 | 30000  |
| 5 | 60000  |

clean\_data

|   | Name  | Domain      | Age | Location  | Salary | Exp |
|---|-------|-------------|-----|-----------|--------|-----|
| 0 | Mike  | Datascience | 34  | Mumbai    | 5000   | 2   |
| 1 | Teddy | Testing     | 45  | Bangalore | 10000  | 3   |

|          | <b>Name</b> | <b>Domain</b> | <b>Age</b> | <b>Location</b> | <b>Salary</b> | <b>Exp</b> |
|----------|-------------|---------------|------------|-----------------|---------------|------------|
| <b>2</b> | Umar        | Dataanalyst   | 50         | Bangalore       | 15000         | 4          |
| <b>3</b> | Jane        | Analytics     | 50         | Hyderbad        | 20000         | 4          |
| <b>4</b> | Uttam       | Statistics    | 67         | Bangalore       | 30000         | 5          |
| <b>5</b> | Kim         | NLP           | 55         | Delhi           | 60000         | 10         |

```
imputation = pd.get_dummies(clean_data)
```

```
imputation
```

|          | <b>Age</b> | <b>Salary</b> | <b>Exp</b> | <b>Name_Jane</b> | <b>Name_Kim</b> | <b>Name_Mike</b> | <b>Name_Teddy</b> |
|----------|------------|---------------|------------|------------------|-----------------|------------------|-------------------|
| <b>0</b> | 34         | 5000          | 2          | False            | False           | True             | False             |
| <b>1</b> | 45         | 10000         | 3          | False            | False           | False            | True              |
| <b>2</b> | 50         | 15000         | 4          | False            | False           | False            | False             |
| <b>3</b> | 50         | 20000         | 4          | True             | False           | False            | False             |
| <b>4</b> | 67         | 30000         | 5          | False            | False           | False            | False             |
| <b>5</b> | 55         | 60000         | 10         | False            | True            | False            | False             |

```
clean_data
```

|          | <b>Name</b> | <b>Domain</b> | <b>Age</b> | <b>Location</b> | <b>Salary</b> | <b>Exp</b> |
|----------|-------------|---------------|------------|-----------------|---------------|------------|
| <b>0</b> | Mike        | Datascience   | 34         | Mumbai          | 5000          | 2          |
| <b>1</b> | Teddy       | Testing       | 45         | Bangalore       | 10000         | 3          |
| <b>2</b> | Umar        | Dataanalyst   | 50         | Bangalore       | 15000         | 4          |
| <b>3</b> | Jane        | Analytics     | 50         | Hyderbad        | 20000         | 4          |
| <b>4</b> | Uttam       | Statistics    | 67         | Bangalore       | 30000         | 5          |
| <b>5</b> | Kim         | NLP           | 55         | Delhi           | 60000         | 10         |

```
imputation
```

|          | <b>Age</b> | <b>Salary</b> | <b>Exp</b> | <b>Name_Jane</b> | <b>Name_Kim</b> | <b>Name_Mike</b> | <b>Name_Teddy</b> |
|----------|------------|---------------|------------|------------------|-----------------|------------------|-------------------|
| <b>0</b> | 34         | 5000          | 2          | False            | False           | True             | False             |
| <b>1</b> | 45         | 10000         | 3          | False            | False           | False            | True              |
| <b>2</b> | 50         | 15000         | 4          | False            | False           | False            | False             |
| <b>3</b> | 50         | 20000         | 4          | True             | False           | False            | False             |
| <b>4</b> | 67         | 30000         | 5          | False            | False           | False            | False             |
| <b>5</b> | 55         | 60000         | 10         | False            | True            | False            | False             |